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Sleep and cardiorespiratory function assessed by a smart bed over 10 weeks post COVID-19 infection

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Inadequate information exists regarding physiological changes post-COVID-19 infection. We used smart beds to record biometric data following COVID-19 infection in nonhospitalized patients. Recordings of daily biometric signals over 14 weeks in 59 COVID-positive participants' homes in 2020 were compared with the same participants' data from 2019. Participants completed a survey of demographic information, health conditions, COVID exposure and testing, and symptom prevalence/subjective severity. Mean age was 47.5 years (standard deviation [SD] 9.5), mean body mass index was 30.1 kg/m^2 (SD 7.1), and 46% were men. During acute infection, 64% exhibited 5–6 h increased sleep duration, 51% had increased movement, and 64% had increased breathing rate (BR). Nearly 34% had paradoxical bradycardia (decreased heart rate by ~10 BPM concomitant with elevated BR and/or fever), with more-severe symptoms. Smart beds can detect physiological changes during COVID-19. A subtype of acute response (paradoxical bradycardia) may predict delay recovery from COVID-19.

The first confirmed cases of COVID-19 in the United States were detected in January 2020, and by the end of the calendar year, the case count had climbed to nearly 20 million[1]. Less than 3% of those with COVID-19 were hospitalized from May 2020 through April 2021[2], with most individuals having a mild infection and quarantining at home[3]. As such, objective data capturing daily viral symptom progression are limited, though subjective reports of symptoms were collected, usually a posteriori, in multiple studies[4,5]. Objective data have largely been extracted from the medical records of older hospitalized patients[6].

The potential of longitudinal biometric signals collected by smart "wearable" devices to document the onset and time course of COVID-19 was recognized early in the pandemic[7,8]. Several studies showed that physiological data including daily activity, sleep, heart rate (HR), and breathing rate (BR), used in conjunction with machine-learning algorithms, may be used to predict and detect COVID-19 infection[9–11].

The profound respiratory and cardiovascular effects of COVID-19 are well documented along with the debilitating impact on sleepiness and fatigue[9,12]. However, there remains a paucity of information regarding the timeframe, dynamics, and relationships of changes in HR, BR, and sleep associated with COVID-19 infection. It is unknown if robust differences between individuals in physiological manifestations detected in the acute phase (defined by the Centers for Disease Control as the 4 weeks following the onset of symptoms) may predict the severity and delayed recovery from COVID-19 illness.

The Sleep Number smart bed, validated against polysomnography, uses high-resolution, full body ballistocardiography to accurately measure movement, BR, and HR[13]. The data are processed to detect sleep and wake states.

The objective of this analysis was to use data collected by the smart bed in the home environment to examine individual patterns of physiological changes associated with acute and postacute COVID-19 infections in patients who tested positive in 2020. Data collected in 2019 from the same individuals were used for comparison.

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Results

Participant demographics and baseline characteristics

The survey was completed by 9370 adults between August and November 2020. Those who did not report the results of a COVID-19 test were excluded, leaving 3546 participants. Only those with a positive COVID-19 test between March 2020 and November 2020 were included ($n = 247$; Fig. 1) in the analysis. Among participants who reported a valid symptom onset date ($n = 190$), 81 were excluded because of a gap of 3 or more days of missing data during the critical period (spanning from 3 days before to 10 days after subjective symptom onset). When queried, 69 of these participants indicated that they had not slept in their smart bed during the critical period. The final analytical sample included 59 participants.

Included participants ($n = 59$) had a mean age of 47.5 ± 9.5 years, a mean body mass index (BMI) of 30.1 ± 7.1 kg/m^2, and 46% (n/n = 27/59) were men (Table 1). Of the cohort, 46% reported no preexisting medical conditions; while high blood pressure was reported by 25%, asthma by 17%, sleep apnea by 10%, and diabetes by 5%. Cardiovascular disease or prior stroke was reported by 1 participant each. Subjective illness severity during the COVID-19 period was reported as mild by 25% of participants, moderate by 64%, and severe/very severe by 10%.

During the baseline periods, there were no significant differences between 2019 and 2020 mean sleep duration, movement, and HR; however, mean BR was slightly higher in 2020 compared to 2019 by an average of 0.22 breaths per minute (Supplementary Appendix, Sect. 3, Supplementary Table 1). All physiological variables recorded in 2020 were highly correlated to those measured in the same participant on the same sleeping surface 1 year earlier (Supplementary Table 1, Supplementary Fig. 1).

Overall mean profiles over the observation period

For all 59 participants, the mean profiles for sleep duration, movement, BR, and HR over the 14-week period surrounding COVID-19 infection in 2020 as compared to habitual conditions in 2019 are illustrated in Fig. 2. Overall, the study population exhibited increases in sleep duration, movement, and BR, while changes in HR appeared more complex. To explore the individual differences in the response to COVID-19 infection, we analyzed each participant's sleep duration, movement, BR, and HR profiles.

Individual differences in responses to COVID-19 infection

Figure 3 illustrates for each physiological variable the impact of separating the participants who did not experience statistically significant changes post time 0 relative to their own baseline variability (lower panels of Fig. 3a–d) from the remainder of the participants who experienced robust changes post-COVID-19 infection.

For sleep duration, a 5–6-h increase in sleep duration following the date of symptom onset was detected in 64% ($n = 38$) of participants (top panel, Fig. 3a). This increase typically peaked 8–9 days later (Supplementary Table 2), when sleep duration averaged more than 12 h. On average for these 38 participants, sleep duration did not return to baseline until the end of the acute phase, at 13.9 days. In the remaining 36% ($n = 21$) of participants,

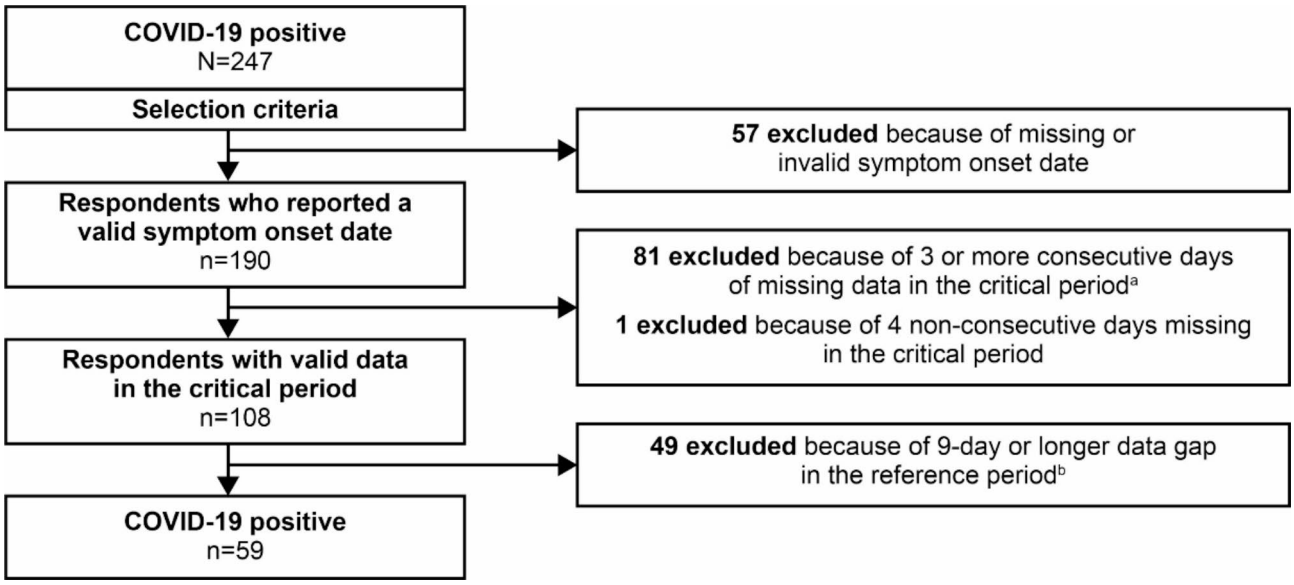

Fig. 1. CONSORT diagram: Participants with a positive COVID-19 test between March 2020 and November 2020 were included in the analysis. Of the survey participants, 190 reported a valid symptom onset date, 81 were excluded because of a gap of 3 or more days of missing data during the critical period, and 49 were excluded because of 9 or more days of missing data in the reference period. In total, 59 participants were included in this study. [a]Critical period (2020): starting 3 days before and ending 10 days after reported date of symptom onset. [b]Reference period (2019): a 14-day period 1 year prior to the critical period in 2020, at the same time of year in 2019. CONSORT, Consolidated Standards of Reporting Trials.

Parameter	COVID-19 severity			
	All	Mild	Moderate	Severe[a]
Participants, no. (%)[b]	59 (100)	15 (25)	38 (64)	6 (10)
Sex, female/male/not reported	30/27/2	7/7/1	19/18/1	4/2/0
Mean age, years (SD)	47.5 (9.5)	46.5 (8.9)	48.4 (9.2)	44.5 (13.6)
Mean weight, kg (SD)	89.4 (23.1)	79.1 (18.8)	90.8 (24.0)	106.0 (17.0)
Mean BMI, kg/m^2 (SD)	30.1 (7.1)	26.7 (5.8)	30.4 (6.7)	36.6 (7.9)
Asthma[c], no. (%)	10 (17)	0 (0)	7 (12)	3 (5)
Cardiovascular disease[c], no. (%)	1 (2)	0 (0)	1 (2)	0 (0)
Diabetes[c], no. (%)	3 (5)	1 (2)	2 (4)	0 (0)
High blood pressure[c], no. (%)	15 (25)	6 (10)	9 (15)	0 (0)
Sleep apnea[c], no. (%)	6 (10)	3 (5)	2 (3)	1 (2)
Stroke[c], no. (%)	1 (2)	0 (0)	1 (2)	0 (0)
No preexisting conditions, no. (%)	27 (46)	6 (10)	18 (31)	3 (5)

Table 1. Participant demographics and characteristics. Categories of subjective COVID-19 severity were defined as follows: mild = symptoms effectively managed at home; moderate = symptoms effectively managed at home; severe/very severe = symptoms requiring hospitalization with/without breathing support (e.g., ventilator or oxygen), respectively. No participants reported intermittent or persistent atrial fibrillation, chronic obstructive pulmonary disease, or heart attack/failure. BMI, body mass index; No., number; SD, standard deviation. [a]"Severe group" included both the severe and very severe groups; [b]group percentages do not equal 100% due to rounding to the nearest whole number for each group; [c]all preexisting conditions were reported by participants and were not clinically defined.

smaller increases of sleep duration were noted but were within the range of baseline variability (lower panel, Fig. 3a). None of the participants exhibited significantly decreased sleep duration during the acute phase of COVID-19 infection.

Movement doubled in 51% ($n = 30$) of participants (Fig. 3b, upper panel). The increase in sleep duration or movement detected in 83% (49/59) of participants indicates an increase in "restless sleep" that persisted for approximately 8 weeks after the onset of symptoms, although individual differences were detected, as described below. In the remaining participants ($n = 29$; lower panel, Fig. 3b), changes in movement were within the range of baseline variability.

An increase in BR was a prominent symptom of the acute phase of infection for 64% ($n = 38$) of participants. Despite being measured at bedrest, peak BR averaged more than 3 breaths per minute higher than at baseline (Fig. 3c, upper panel; Supplementary Table 2). Among the remaining participants, (36%, $n = 21$; Fig. 3c, lower panel), changes in BR were within the range of baseline variability.

For HR measures, 3 patterns emerged from analysis of individual profiles (Fig. 3d): (1) in 25% ($n = 15$) of participants, increased HR during the acute phase was detected, with variable timing relative to the onset of symptoms (Fig. 3d, upper panel; Supplementary Table 2); (2) unexpectedly, 34% ($n = 20$) of HR profiles were characterized by a progressive decrease in HR, beginning several days after symptom onset and lasting for more than 2 weeks (Fig. 3d, middle panel), with the decrease in HR averaging nearly 10 beats per minute (bpm) (Supplementary Table 2) and, for 20/59 of participants, this HR decrease occurred concomitantly with a significant elevation of BR—consistent with "paradoxical bradycardia" as, under normal conditions, an elevation of BR is associated with a faster, rather than slower, HR; (3) the remaining 41% ($n = 24$) of participants did not exhibit significant changes relative to variability in baseline HR (Fig. 3d, lower panel). Figure 3d further illustrates that the cohort exhibiting decreasing, rather than increasing, HR during the acute phase had higher mean baseline HR (2020 baseline mean HR prior to acute-phase decrease was 66.4 ± 5.1 bpm versus 60.3 ± 8.4 bpm prior to acute-phase increase; $P = 0.0134$).

Overall, only 10% ($n = 6$) of participants did not exhibit statistically significant ($P > 0.05$) changes in any of the 4 physiological signals, while 17% ($n = 10$) exhibited robust changes in all 4 metrics. Almost 40% of participants experienced a significant change in 3 of the 4 recorded metrics.

We also explored associations between subtypes of sleep, movement, BR, and HR profiles and the reported number and severity of individual symptoms. Out of 11 symptoms, sleepiness/fatigue was most frequently reported (85%, $n = 50$); Supplementary Fig. 2a), while nausea/vomiting was reported by fewer than 20% ($n = 11$). Symptoms with lower prevalence were not associated with lower severity (Supplementary Fig. 2b). In participants showing increases in BR ($n = 38$), the only trending symptom association was with congestion/runny nose ($P = 0.054$). Among participants with a decrease in HR ($n = 20$), associations between fever/chills, cough, and diarrhea were found ($P < 0.05$ for all). An acute decrease in HR was concomitant with increased BR and/or fever in 18 of 20 participants, suggesting that the term "paradoxical bradycardia" is appropriate to describe this type of HR response to COVID.

Symptom severity of subtype profiles and subtypes of cohorts with changing HR

Multivariate linear regression was used to explore the relationship between the total severity score of the 11 symptoms queried in the survey and the subtype of profile for each of the 4 metrics. This analysis revealed that

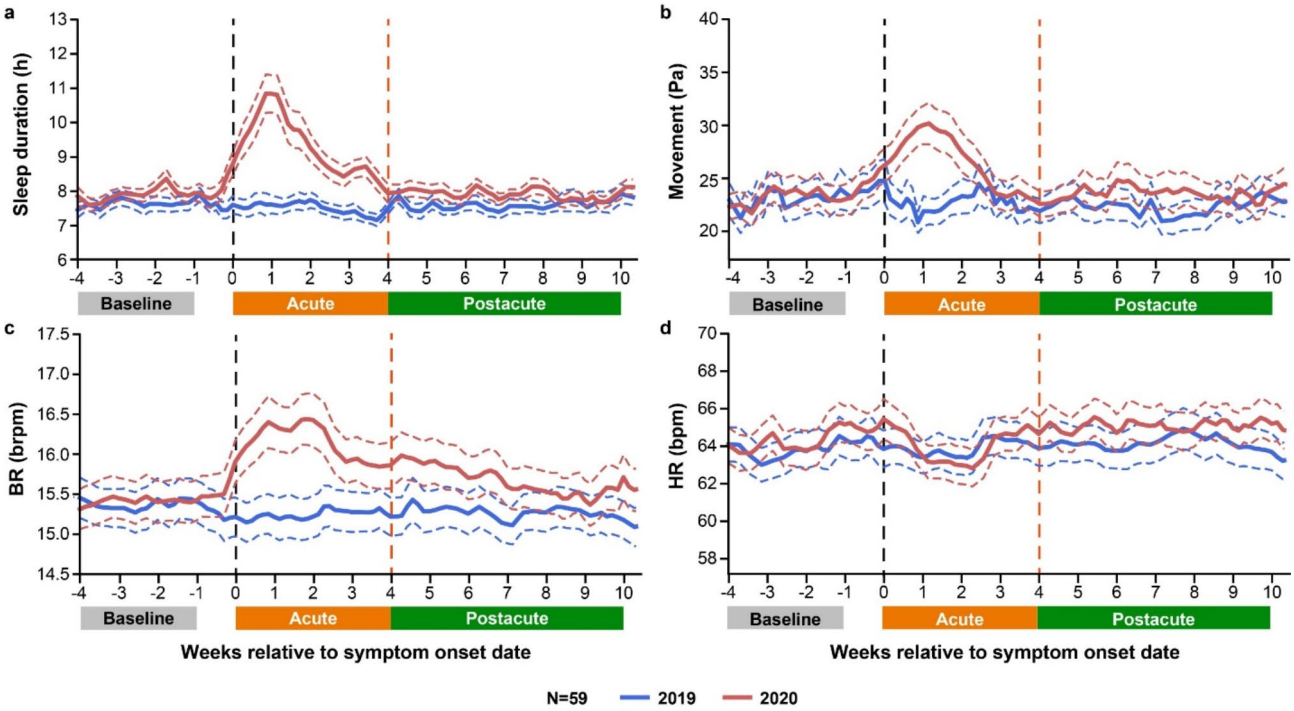

Fig. 2. Parts a–d: differences in mean profiles of (a) sleep duration, (b) movement, (c) BR and (d) HR in response to COVID-19 infection in 2020 versus 2019. For all 59 participants, the mean profiles for (a) sleep duration, (b) movement, (c) BR and (d) HR over the 14-week period surrounding COVID-19 infection in 2020 as compared to habitual conditions in 2019 are illustrated. Mean (solid lines) values for each metric in 2020 (**red**) and the corresponding period in 2019 (**blue**) for the 59 participants are shown. For both 2019 and 2020, the mean ± SEM is represented by dashed lines below and above the mean. All data are referenced relative to time 0 (shown as a vertical **black** dashed line), the date of onset of symptoms in 2020, and the corresponding date in 2019. The baseline period was − 4 to − 1 weeks (indicated by the **gray** horizontal bar), the acute phase of infection was time 0 to + 4 weeks (indicated by the **orange** horizontal bar), and the postacute phase was weeks + 5 to + 10 (indicated by the **green** horizontal bar) as represented on the abscissa. The vertical **red** dashed line indicates the end of the acute phase and beginning of the postacute phase. bpm, beats per minute; BR, breathing rate; brpm, breaths per minute; h, hour; HR, heart rate; Pa, Pascal; SEM, standard error of the mean.

the total symptom severity score was associated with the subtype of HR profile, where no change in HR was associated with lower severity ($P = 0.015$) and a decrease in HR trended toward association with higher severity ($P = 0.071$). Increases in sleep duration, movement, and BR were not identified as significant drivers of symptom severity (Supplementary Fig. 3).

The analysis of the effects of sleep duration, motion level, heart rate and breathing rate profile subtype during the acute phase on post-acute phase profiles using linear mixed effects model (Supplementary Table 3) shows significant effects of the heart rate subtype profile on sleep duration and breathing rate during the post-acute phase.

The profiles of HR, sleep duration, and BR during the 6-week post-acute phase are illustrated for each subtype of HR acute response to infection in Fig. 4a–c (top: no change in HR, middle: decrease in HR, bottom: increase in HR). The "paradoxical bradycardia" cohort ($n = 20$) was uniquely characterized by significant elevations of both sleep duration and BR persisting until the end of the 10-week post-acute phase. In contrast, despite their drop in HR during the acute phase, the "bradycardia cohort" recovered higher BR levels in the post-acute phase, similar to the levels observed under baseline conditions.

Overall, paradoxical bradycardia during the acute phase of COVID-19 infection was associated with more-severe symptoms and more-prolonged elevations of sleep duration and BR in the post-acute phase. No reported preexisting health conditions were significantly associated with paradoxical bradycardia during the acute phase of infection.

Discussion

Our study, which spanned 14 weeks and encompassed the acute and post-acute phases of COVID-19 infection in 2020, underscores the feasibility of utilizing smart beds to capture and analyze physiological signals during sleep. These data are a unique source of longitudinal health data for potentially predicting and monitoring the development of symptoms associated with respiratory diseases, including COVID-19. Comparing physiological signals recorded in the same individuals at the same time of year in 2019 provided a valuable context for evaluating the changes associated with infection. Importantly, all data were collected within the real-world settings

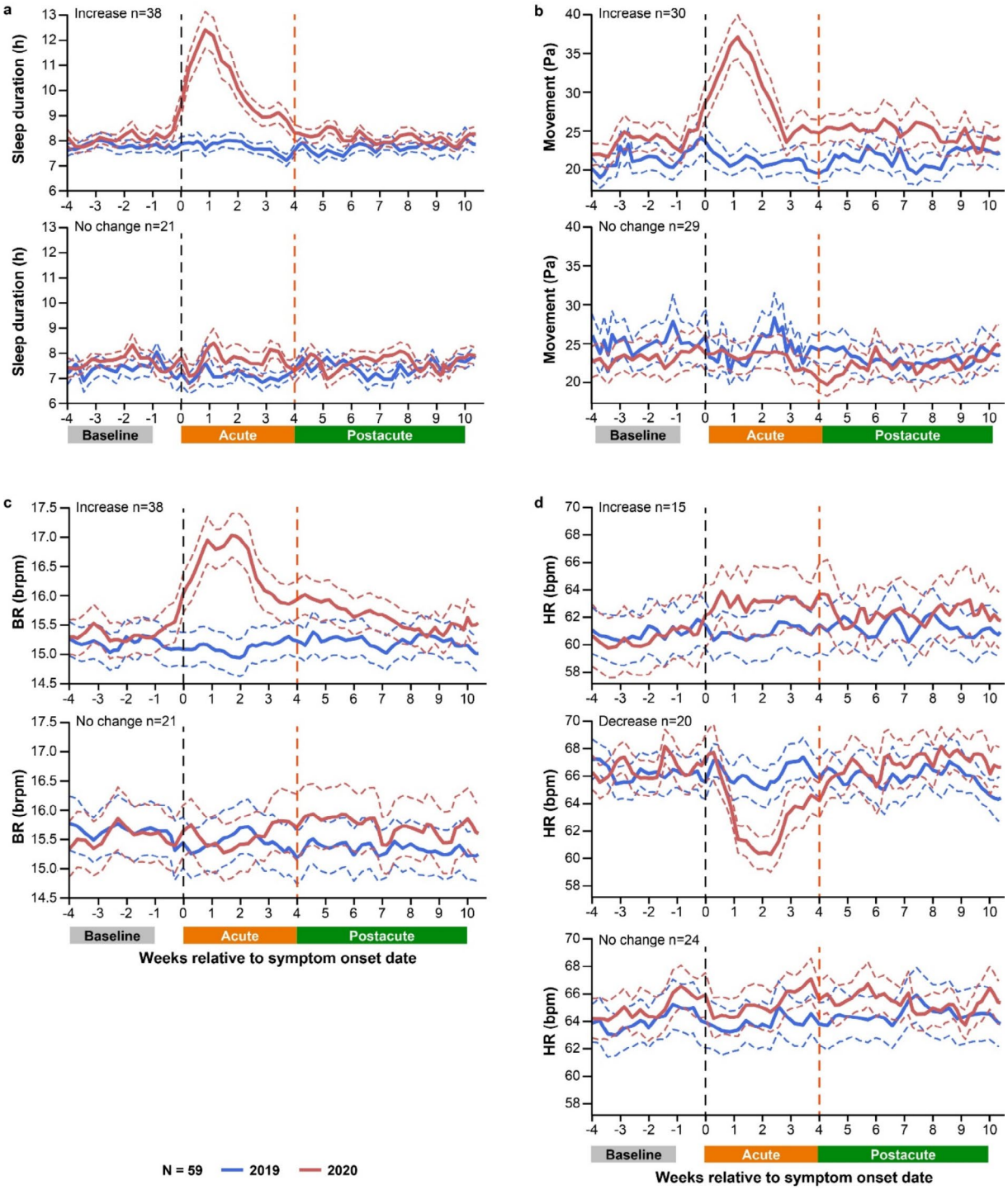

Fig. 3. Parts a–d: subtypes of mean profiles of (a) sleep duration, (b) movement, (c) BR, and (d) HR. Participants that did not experience significant changes after time zero of COVID-19 infection were separated from participants with robust changes in each physiological variable. Mean profiles for participants with significant changes or no significant changes in (a) sleep duration, (b) movement, (c) BR, and (d) HR are illustrated. Mean (solid lines) and ± SEM (dotted lines) values for each metric in 2020 (**red**) and the corresponding period in 2019 (**blue**) for subgroups of participants are shown. The baseline period was − 4 to − 1 weeks (indicated by the **gray** horizontal bar), the acute phase of infection was time 0 to + 4 weeks (indicated by the **orange** horizontal bar) and the postacute phase was weeks + 5 to + 10 (indicated by the **green** horizontal bar). The vertical **black** dashed line indicates subjective symptom onset date. The vertical **red** dashed line indicates the end of the acute phase and beginning of the postacute phase. All data are referenced relative to time 0, the date of onset of symptoms in 2020 and the corresponding date in 2019. bpm, beats per minute; BR, breathing rate; brpm, breaths per minute; h, hour; HR, heart rate; Pa, Pascal; SEM, standard error of the mean.

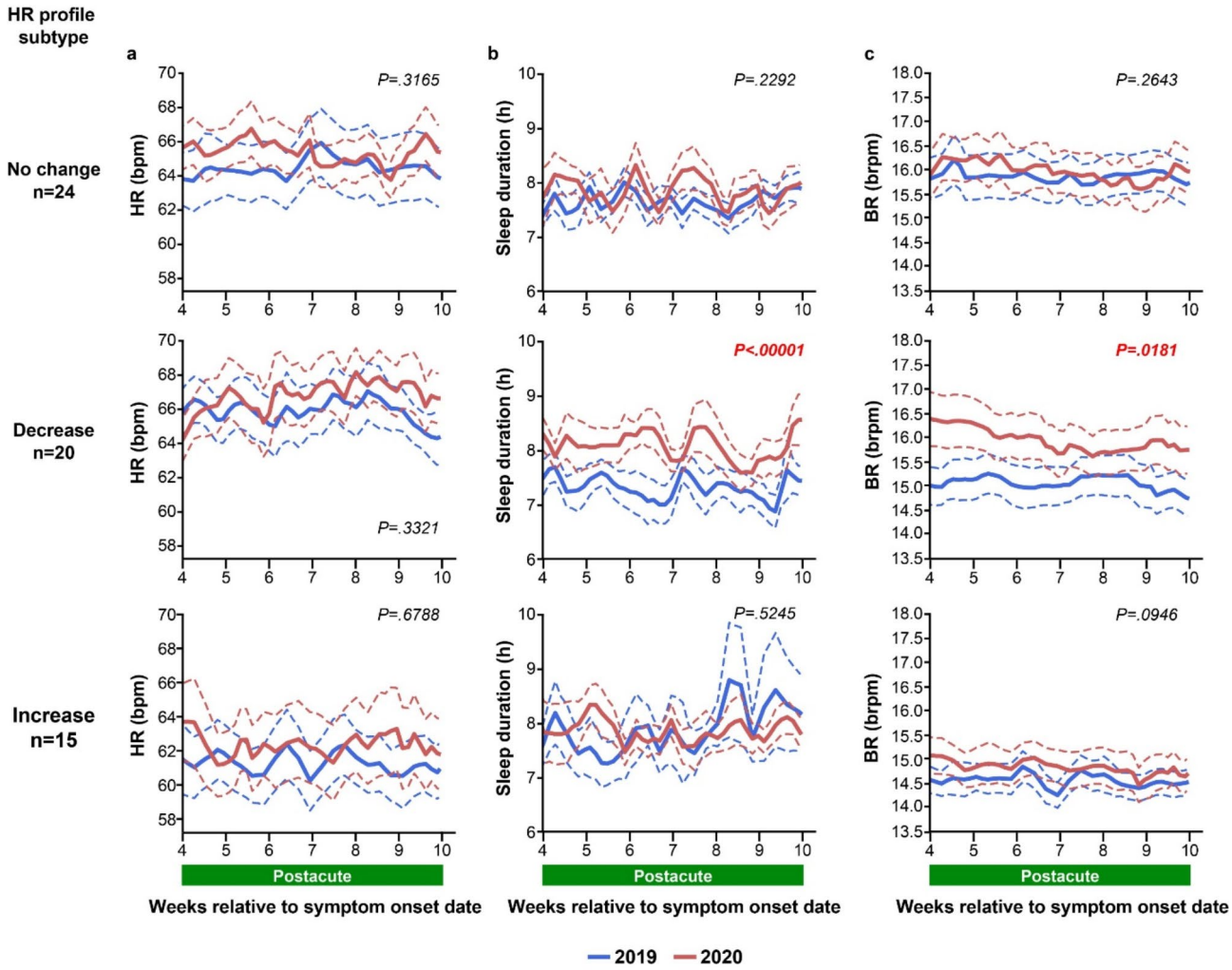

Fig. 4. Parts a–c: postacute phase mean profiles from 3 subtypes of HR for (a) HR, (b) sleep duration, and (c) BR. The profiles of HR, sleep duration, and BR during the 6-week postacute phase are illustrated for each subtype (no change, decrease, or increase) of acute HR response to infection. Mean (solid lines) and ± SEM (dotted lines) values for HR, sleep duration, and BR in 2020 (**red**) and the corresponding period in 2019 (**blue**) for subgroups of participants are shown. The postacute phase was weeks + 5 to + 10 (indicated by the **green** horizontal bar). All data are referenced relative to time 0, the date of onset of symptoms in 2020 and the corresponding date in 2019. bpm, beats per minute; BR, breathing rate; brpm, breaths per minute; h, hour; HR, heart rate; SEM, standard error of the mean.

of participants' homes, ensuring that our findings accurately reflect physiological and symptomatic changes experienced during infection. The confined environment of the bed further controlled variability in physical activity that typically characterizes data from wearable monitors (i.e., activity trackers). Lastly, collection of physiological variables during sleep by the smart bed was previously validated against polysomnography[13] and mean overnight HR and BR were highly correlated with PSG-derived metrics ($r > 0.90$). Our data highlight the distinct characteristics of individual COVID-19 symptom manifestations and associated clinical characteristics, which may inform disease trajectories and guide targeted interventions.

Studies using wearable technologies revealed early physiological changes associated with COVID-19 infection, sometimes even preceding symptom onset. Research utilizing the Oura ring detected nightly changes in skin temperature, BR, HR, heart rate variability (HRV), and sleep efficiency during the month preceding infection or vaccination, showing physiological responses as early as 2.5 days before reported symptom onset[10]. Another study utilizing Fitbit data predicted COVID-19 onset and severity through the analysis of BR, HR, and HRV[11]. These authors reported significant changes in BR, returning to baseline after day 10, and HR variations that initially increased during acute sickness then decreased below baseline values from days 7–21. Quer and colleagues (2021) reported that sleep and activity data obtained from various wearable devices like Apple, Fitbit, and Google could effectively differentiate between COVID-19 positive and negative individuals[9], highlighting the utility of wearables for longitudinally monitoring physiological data. To our knowledge, analyses of individual differences in physiological responses to COVID-19 detected by wearables in affected individuals have not yet been reported.

Our analysis revealed that almost 40% of participants experienced significant changes in 3 of the 4 physiological metrics assessed, with key variations in sleep duration, BR, and HR during the acute phase of COVID-19. Large increases in sleep duration (considered "restless sleep" due to concomitant increases in movement), and elevations of BR peaked around days 8–10 following COVID-19 symptom onset. While HR changes remained within the range of habitual variability in 40% of the participants, a large and extended HR decrease, averaging nearly 10 bpm, was initiated towards the end of the first week in about one third of participants. In these participants, increased sleep duration and BR persisted until the end of the analytical period, indicating the potential utility of these metrics as indicators of longer lasting symptoms. The decline in HR, a relatively late-occurring characteristic, could potentially be preventable with medical intervention. Our findings, along with those from previously published studies[9–12], suggest that using personal metric-recording wearables (such wearables able to monitor movement, photoplethysmography, and temperature) and nearables may offer valuable insights into the severity and longevity of an illness, potentially enabling timely interventions.

Cardiac manifestations of COVID-19 and other coronaviruses are well-documented[14–18]. For COVID-19, sinus bradycardia is one common arrhythmia, characterized by a slower-than-normal HR[15]. In a retrospective analysis of the medical records of 1053 patients diagnosed with COVID-19, absolute bradycardia (HR < 60) and profound bradycardia (HR < 50) were found in 25% and 13% of patients, respectively[16]. While bradycardia has been documented in hospitalized patients with COVID-19, its occurrence in less-severe cases where individuals recovered at home remains underreported[19]. In our study, a discernible pattern of HR changes during the acute phase of COVID-19 was identified. This pattern was characterized by a notable HR decrease of up to 10 bpm over approximately 10 days. This decrease was observed in 34% of relatively young and previously healthy participants and coincided with an increase in BR and reports of fever/chills. This observation contrasted with the faster HR that would be reasonably anticipated, leading us to use the term "paradoxical bradycardia" to describe this group. In contrast to a prior report[14], we noted that some participants with paradoxical bradycardia had extended duration of noncardiac symptoms, including elevated BR and increased sleep duration. These factors may serve as potential indicators of "long COVID." Notably, this subgroup had distinct characteristics, including a slightly older age, higher BMI, and elevated HR under baseline conditions in both 2019 and 2020 compared with other participants. During their COVID-19 infection, these participants also reported symptoms such as fever/chills, cough, and diarrhea. Our study demonstrates the unique ability of the smart bed to serve as an early indicator in a home environment for those potentially in need of further medical assessment.

Our findings, together with our prior research, underscore the potential health and wellness value of smart bed technology for monitoring sleep metrics informing about influenza-like symptoms[20], the prevalence of influenza depending on chronic sleep conditions such as insomnia[21], and the impact of sleep behaviors such as regularity and wake/sleep timing on sleep quality[22]. In addition, the combination of smart bed-based metrics with modern machine learning techniques enabled the prediction of chronic insomnia[23] which can support preventive health strategies. These capabilities suggest that smart beds may provide valuable, non-intrusive insights into sleep health, with practical implications for early detection of sleep-related disorders and for supporting interventions that enhance sleep quality in the general population.

This study has several limitations that may impact the interpretation and generalizability of the findings. The sample size was relatively small but this was partly compensated for by the reliability of the metrics collected by the smart bed and the availability of the within-subject reference period in 2019[13]. Our analysis implemented stringent criteria to detect significant changes in habitual physiological variability. Thus, the study did not quantify "minor" changes in physiological metrics, potentially limiting the understanding of more subtle changes. Data collection occurred during the early stages of the pandemic, which poses challenges due to evolving viral variants and changing symptom patterns. Additionally, subjective symptom reporting by participants could have introduced bias, inaccuracy, and variability, impacting the reliability and objectivity of the data. Similar to actigraphy, the smart bed[13] demonstrates high sensitivity to detect sleep (= 0.94) and moderate specificity to detect wake (= 0.48), suggesting a potential overestimation of sleep duration. Further, since the changes in sleep metrics that we observed in this research are associated with general immune responses, these could represent several illnesses, respiratory or otherwise[24]. Finally, the study participant pool was limited to a subgroup of COVID-19-positive individuals with complete records. The exclusion of those without an end date of symptoms may have removed some individuals with "long COVID" from the study. In the future, there is a need for additional data collection to confirm and extend the present findings.

Conclusions

In summary, unobtrusive monitoring of physiological variables during sleep using smart beds can reveal important insights into individuals' cardiovascular health indicators, sleep patterns, and objective symptoms during COVID-19 illness. Future directions include the analysis of larger samples over a longer recording period consistent with "long COVID," as well as the development of a predictive model based on the subtypes of physiological profiles reported in this study. We believe that signals collected by a smart bed have the potential to provide an early indicator of infection, symptom onset and progression, thereby contributing to improved patient care and public-health strategies.

Methods

This research was conducted in accordance with US federal guidelines (Common Rule) regulating the protection of human subjects participating in biomedical and behavioral research. The protocol describing this research # P2020/03001 NEIRB 17-1323071-1 was approved by the privately owned WCG New England Institutional Review Board (IRB).

Participants

Participants were users of the Sleep Number smart bed. Data were collected daily. Informed consent was obtained from all participants using the Sleep IQ app (Sleep Number, Minneapolis, MN) on the participant's smart phone, computer, or other electronic device of their choice. Participants gave access to nightly physiological measurements, which were electronically stored in the cloud after being anonymized. Participants completed an IRB approved (NEIRB 17-1323071-1) survey between August 2020 and November 2020.

The online administered survey (Supplementary Appendix, Sect. 1) via email invitation, included questions about participants' demographics, preexisting health conditions, and COVID-19–specific exposure, testing, diagnosis, subjective evaluation of illness severity, and presence and severity of 11 key symptoms.

Participants who reported having tested positive for COVID-19 between March 2020 and November 2020 and who completed the survey were eligible. Additional inclusion criteria were (Fig. 1): reporting start and end dates of the subjectively defined symptomatic period (month/day) corresponding to their first or most recent positive test; the start date was before the end date; no gaps in stored data exceeding 3 days or more in the critical period; and fewer than 4 missing sleep sessions in the critical period. For each participant, we defined time 0 as the day of onset of subjective symptoms, and the "critical period" as starting 3 days before and ending 10 days after time 0. We defined the "reference period" as a 14-day period in 2019 corresponding to the critical period in 2020, with 2019 time 0 on the same day of the week as time 0 in 2020 and with 2019 data available within a window of ± 14 calendar days from the critical period in 2020.

Severity of COVID-19 disease and symptoms were subjectively reported and analyzed as outlined in Supplementary Appendix, Sect. 2, Supplementary Methods.

Data processing

Anomalous data detection and correction

Anomalous data were identified by considering the multivariate distribution of HR, BR, sleep duration, and motion. For each time point (i.e., day), an anomaly score was calculated using the standard 'local outlier factor' algorithm[25], which estimates the multivariate distribution around each point. The anomaly score was < 1 when the datapoint was near the mode of the distribution, = 1 when the datapoint coincided with the mode, and > 1 as the datapoint moved farther away from the mode. The metrics of days with an anomaly score exceeding 1 were corrected using an average weighted by the anomalous score between an interpolated value using the data surrounding the anomalous day and the metrics of the anomalous day. Thus, the more anomalous a data point was, the higher the importance of the interpolated data to obtain the corrected value.

Missing data interpolation

Missing data were differentially handled depending on the number of contiguous missing days and when they occurred with respect to symptom onset. For the 2020 data, a maximum of 2 contiguous missing sleep sessions were interpolated in the two-week period following symptom onset. Outside this two-week period, we interpolated a maximum of seven contiguous missing sleep sessions. The interpolation was performed causally with exponentially decaying weights applied to a 7-day window.

For the 2019 data, interpolation was performed using a 14-day window centered around the date to be interpolated. The details of this procedure are illustrated in Supplementary Fig. 1. Overall, for the 59 participants, the mean percentage of interpolated data was 8.29% (median: 7.76%, standard deviation [SD] 5.54%, interquartile range: 3.67–11.6%).

Data curation and organization

All individual biometric profiles were smoothed by a widely used robust locally weighted regression (LOWESS) procedure to minimize the impact of sporadic up-and-down spikes on the longer-term temporal variations of interest[26]. We used a regression window of 5 days, consistent with the time-frame of the development of infection-related symptoms. All subsequent analyses were based on the smoothed profiles.

For each participant, the 14 weeks of physiological data were organized in nightly sessions, each corresponding to 1 night of bedrest/sleep. For each nightly session, mean sleep duration, movement, BR, and HR were derived from the signals recorded by the smart bed. The analytical period spanned 14 weeks and was subdivided into a 3-week baseline period (from week − 4 to week − 1 prior to time 0), a 4-week acute phase (from time 0 to week + 4), and a 6-week postacute phase (from week + 5 to week + 10). The corresponding time intervals were analyzed for 2019 and 2020. Data collected during week − 1 were not considered part of the baseline period as changes in physiological variables were detected by the smart bed a few days prior to the reported date of symptom onset.

Data analysis

Thresholds for sleep duration, movement, BR, and HR were established to determine when changes in each metric after COVID-19 infection in 2020 exceeded the mean day-to-day individual variability in the baseline periods examined in 2019 and 2020. As the distribution of each metric at baseline approximated a Gaussian curve, we used the mean ± 1.96 standard deviation (SD) of the data collected under baseline conditions as the upper or lower threshold for a significant change. Significant changes ($P < 0.05$) in each metric were identified algorithmically, then borderline cases ($P = 0.05$) were confirmed by visual examination. In rare instances (in 5 out of 59 cases) of a discrepancy, a decision was reached by author consensus.

Definitions of baseline levels and significant changes from baseline

For each physiological metric, baseline mean level and day to day variability were defined using individual means and SD of values recorded during the baseline period in 2020 and during the corresponding period in 2019.

- 2020 mean and SD were calculated over the 3-week period preceding day − 7 in 2020.
- 2019 mean and SD were calculated over the 3-week reference period.
- The mean baseline was estimated at 0.5 $*(\text{mean}_{2020} + \text{mean}_{2019})$ and the mean SD was estimated as $0.5*(\text{SD}_{2020} + \text{SD}_{2019})$.

Supplementary Table 1 lists the 2020 and 2019 means and coefficient of variation (CV) and shows the year-to-year correlations. These observations are illustrated in Supplementary Fig. 2. For each metric and each participant, the threshold was defined as the 95% confidence limit (1.96 SD) of the distribution of individual baseline values. Onset was defined as index of the first day, relative to the symptom onset, when a significant increase or decrease was identified for a given metric. Offset was defined as index of the first day, occurring at a later calendar date than the onset, where the metric fell back in the interval (baseline − 1.96 SD; baseline + 1.96 SD).

A significant increase for a given sleep metric was identified when the value exceeded the baseline + 1.96 SD. A significant decrease was identified when the value was lower than the baseline − 1.96 SD. The definitions for "significant decline" were symmetric.

Effect of profile type on metrics in the postacute phase

The relationship between the profile subtype during the acute phase (i.e., breathing rate increase/no change, sleep duration increase/no change, movement level increase/no change, heart rate increase/decrease/no change) and the profiles of sleep duration, motion level, and breathing rate in the post-acute phase were explored using linear mixed effect models (LMEMs).

LMEMs used the profile type, the year, as fixed factors and the subject identifier as random factor.

Subjective disease and symptom severity

Overall severity of COVID-19 disease was subjectively reported as one of the following: no symptoms; mild (effectively managed at home); moderate (effectively managed at home); severe (required brief hospitalization); or very severe (required hospitalization and breathing support). For this study, participants reporting severe or very severe COVID-19 symptoms were combined (n/n: 6/59). The survey asked the participants about the presence (yes/no) and the severity (scored on a scale of 1 to 10 for each) of 11 symptoms previously reported by COVID-19 patients, including sleepiness/fatigue, muscle/body aches, headache, fever/chills, cough, new loss of taste/smell, congestion/running nose, shortness of breath, diarrhea, nausea/vomiting and sore throat. Two participants reported additional symptoms of "numbness" and "pinkeye".

The mean severity score of each symptom across all participants and the sum of the severity scores for all reported symptoms for each participant were calculated (Supplementary Fig. 3). Multivariate linear regression was used to explore the relationship between the total score of severity of the 11 symptoms and the subtype of profile for each of the 4 metrics. Results are shown in Supplementary Fig. 4.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to limitations in the consent form that was signed by participants in the study. However, summary unidentified data are available from the corresponding author on reasonable request.

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Concept and design: EVC, VKS.Acquisition, analysis, or interpretation of data: EVC, GG-M, DG, VKS, SDeF. Drafting of the manuscript: GG-M, MA, VKS, FM, RM, and EVC.Statistical analysis: EVC, GG-M, DGSupervision: RM, and FMCritical review of the manuscript for important intellectual content: All authors.

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Competing interests
Authors GG-M and FM are employees of Sleep Number Labs.Authors SDeF, MSA, and RM are employees of Sleep Number Corporation. Authors GG-M, DG, FM, RM, MSA, SDeF, EVC, VS are inventors on a smart bed-related patent applications (WIPO Patent Application WO/2022/120165; filed on November 16, 2023; Title: Bed having features for automatic sensing of illness state such as postacute sequalae of COVID-19 [PASC or Long COVID]). Author DG is an employee of GlobalLogic, a consulting company under contract with Sleep Number Corporation. Author VKS consulted for ApniMed, Know Labs, Lilly, Jazz Pharmaceuticals, and Zoll, and is on the Sleep Number Scientific Advisory Board. Author EVC is a consultant for the Sleep Number Corporation, for Calibrate Health, Inc, and for the University of Illinois at Chicago. She is the recipient of an investigator-initiated research grant on "Circadian Misalignment in Adrenal Insufficiency" from the Takeda Pharmaceutical Company and is a member of the Scientific Advisory Board of the Sleep Number Corporation.

Additional information
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